Cache-Aided Linear Computation

Xiaohong Cai, Ruze Zhang, Laigang Guo, and Raymond W. Yeung

***Abstract*—We study cache-aided linear computation in distributed systems where each source generates and stores its own data. An external receiver requests a linear combination of these data, and the sources use information cached before the request is known to reduce the communication needed to recover the requested result. In our model, each of the N sources stores its own independent file over a finite field and has an additional cache of normalized size M. The receiver must recover any requested linear combination with zero error from the messages sent by the sources over separate noiseless links. We establish a lower bound on the worst-case communication rate over any finite field by analyzing several requests under the same cache placement. This bound determines the optimal rate when no additional cache is available. For binary files, we propose parity-aided placement (PAP), a coded-placement scheme that combines cached parities with uncoded symbols. For every $N \geq 3$, we characterize its communication rate for each request and optimize its placement parameters to obtain a closed-form worst-case rate. The resulting achievable rate meets the lower bound for $0 \leq M \leq 1/(2N-1)$ and $M \geq (N-1)/N$, establishing the optimal memory–rate tradeoff in these two ranges.**

***Index Terms*—Cache-aided computation, coded placement, distributed computation, linear computation, memory–rate tradeoff**

I. Introduction

In distributed systems with locally generated data, different source nodes retain their own data, while an external receiver may request a function of the data held by multiple sources. For example, Earth observation satellites can share observation data for processing in orbit [1], and multiple aerial sensing nodes can send their observations to a ground station for data fusion through an unmanned aerial vehicle (UAV) relay [2]. When the receiver needs only a function of the source data, a small result does not necessarily mean that little communication is required. The receiver may still need separate transmissions from many sources, whose total size can far exceed that of the result. Exchanging and caching information before the requested function is specified can change the information available at each source and reduce the communication needed for subsequent computation. The challenge is to select cache contents that support different future requests without knowing which function will be requested.

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Caching can reduce the transmission load by making information available before requests arrive. In the classical coded-caching model, a server holds the entire file library, and receivers cache information to reduce the communication required for subsequent file requests over a shared link [3], [4]. Subsequent work established bounds for general placement [5] and developed coded placements that cache linear combinations of file segments [6]. Cache-aided scalar and general linear function retrieval extend this setting to requests for linear functions of the library files, with the entire library still held by a central server and caches located at the receivers [7], [8]. Device-to-device caching enables users to retrieve files from one another's caches [9], [10]. Coding can also reduce communication in distributed computing [11], [12].

Network function computation studies the recovery of functions of distributed source messages over a network. Previous work established cut-set bounds for network function computation [13], algebraic conditions for computing linear functions [14], and improved capacity bounds for general target functions and network topologies [15]. For the binary arithmetic sum, the zero-error compression capacity has been characterized under different source-observation patterns at two encoders [16].

We study how caching at the sources can reduce the communication required for linear function computation. In our model, each of the N sources retains its own file and has an additional cache of normalized size M. The caching functions are fixed before the request is revealed and must support every linear combination requested by an external receiver. We seek to characterize the tradeoff between cache size and the worst-case total communication required for zero-error recovery. Our main results are as follows.

We establish a converse bound for an arbitrary finite field. For any $m \times N$ query matrix A, the bound depends on the average rank of its $m \times n$ column submatrices. Maximizing this quantity over A for each pair (m, n) gives

$$R(\mathbb{F}_q, N, M) \geq \max_{1 \leq m,\, n \leq N} \frac{N}{mn} \big(\widehat{\mathrm{rank}}_{m,n,N} - (N-n)M \big). \tag{1}$$

For the binary case, we introduce *parity-aided placement* (PAP), which combines cached parities with uncoded symbols. For every $N \geq 3$, we derive its communication rate for each request support size and optimize the placement parameters to obtain a closed-form worst-case rate, asymptotically achievable for every M. In the small-cache regime, PAP meets the $m = n = 1$ instance of (1) and gives

$$R(\mathbb{F}_2, N, M) = N - N(N-1)M, \quad 0 \leq M \leq \frac{1}{2N-1}. \tag{2}$$

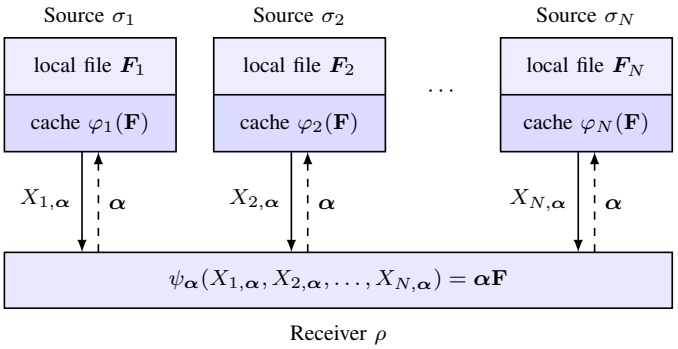


Fig. 1. The model of cache-aided linear computation $(\mathbb{F}_q, N, M)$.

PAP also achieves the optimal rate $R(\mathbb{F}_2, N, M) = 1$ for $M \geq (N-1)/N$, thereby determining the optimal memory–rate tradeoff in these two regimes.

II. Model of Cache-Aided Linear Computation

We consider a *cache-aided linear computation model* $(\mathbb{F}_q, N, M)$ as depicted in Fig. 1. In the model, there are N *sources* $\sigma_1, \sigma_2, \cdots, \sigma_N$ and a *receiver* ρ, where each source is connected to the receiver by a noiseless link. For each $i \in [N] \triangleq \{1, 2, \cdots, N\}$, let the *file* generated by the source σ_i be a random L-dimensional row vector $\boldsymbol{F}_i = (F_{i,1}, F_{i,2}, \cdots, F_{i,L})$ uniformly distributed over $\mathbb{F}_q^L$,[1] where $\mathbb{F}_q$ is a finite field of size q and L is a positive integer referred to as the *file length*. All files $\boldsymbol{F}_1, \boldsymbol{F}_2, \cdots, \boldsymbol{F}_N$ are assumed to be mutually independent. Further, let $\mathbf{F} \triangleq \left[\boldsymbol{F}_1^{\mathsf{T}}\ \boldsymbol{F}_2^{\mathsf{T}}\ \cdots\ \boldsymbol{F}_N^{\mathsf{T}}\right]^{\mathsf{T}}$, which is referred to as the *file matrix*.

Furthermore, due to prior data exchanges, prefetching, or overhearing via the broadcast nature of the wireless channel, the sources have access to some side information about other files. More precisely, for each $i \in [N]$, the source σ_i can store an arbitrary (possibly nonlinear) function of the file matrix $\mathbf{F}$ as the *cached content*, and the cached content takes at most q^{ML} possible values. Accordingly, we refer to ML as the *cache size*. Here, M is a nonnegative real number called the *normalized cache size*. The caching functions are fixed before the coefficient vector $\boldsymbol{\alpha}$ is revealed. For each *coefficient vector* $\boldsymbol{\alpha} \triangleq (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_q^N$, the receiver ρ is required to compute the linear function $\boldsymbol{\alpha}\mathbf{F} = \sum_{i=1}^N \alpha_i \boldsymbol{F}_i$ with zero error through the following steps:

1) the receiver ρ first broadcasts the coefficient vector $\boldsymbol{\alpha}$ to all source nodes;
2) each source node σ_i generates and transmits an encoded message, say $X_{i,\boldsymbol{\alpha}}$, based on its local file, its cached content and the coefficient vector $\boldsymbol{\alpha}$;
3) after receiving all transmitted messages, the receiver ρ computes $\boldsymbol{\alpha}\mathbf{F}$ with zero error.

Now, we formally define an *L-dimensional (linear-computation) code* for the model $(\mathbb{F}_q, N, M)$ such that for each coefficient vector $\boldsymbol{\alpha} \in \mathbb{F}_q^N$, the receiver ρ can compute the linear function $\boldsymbol{\alpha}\mathbf{F}$ of the L-dimensional files $\boldsymbol{F}_1, \boldsymbol{F}_2, \cdots, \boldsymbol{F}_N$ with zero error. To be specific, an L-dimensional code for the model $(\mathbb{F}_q, N, M)$ consists of

- a *caching function* φ_i for generating the cache content at every source node σ_i $\varphi_i : \mathbb{F}_q^{N\times L} \to \operatorname{Im}\varphi_i$ such that the cached content takes at most q^{ML} possible values, i.e., $|\operatorname{Im}\varphi_i| \leq q^{ML}$;[2]
- an *encoding function* $\phi_{i,\boldsymbol{\alpha}}$ for each $\boldsymbol{\alpha} \in \mathbb{F}_q^N$ at every source node σ_i $\phi_{i,\boldsymbol{\alpha}} : \mathbb{F}_q^L \times \operatorname{Im}\varphi_i \to \operatorname{Im}\phi_{i,\boldsymbol{\alpha}}$, which is used to generate a message transmitted from the source node σ_i to the receiver ρ based on the local file $\boldsymbol{F}_i$, the cached content $\varphi_i(\mathbf{F})$ and the coefficient vector $\boldsymbol{\alpha}$;
- a *decoding function* $\psi_{\boldsymbol{\alpha}}$ for each $\boldsymbol{\alpha} \in \mathbb{F}_q^N$ at the receiver ρ, given by $\psi_{\boldsymbol{\alpha}} : \prod_{i=1}^N \operatorname{Im}\phi_{i,\boldsymbol{\alpha}} \to \mathbb{F}_q^L$, which is used to compute the linear function $\boldsymbol{\alpha}\mathbf{F}$ based on the messages received from all the N sources.

We write this code as $\mathbf{C} = \left\{\varphi_i, \phi_{i,\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} :\ i \in [N], \boldsymbol{\alpha} \in \mathbb{F}_q^N\right\}$, and it is called *admissible* if

$$\psi_{\boldsymbol{\alpha}}\Big(\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i,\ \varphi_i(\mathbf{F})\big) :\ i \in [N]\Big) = \boldsymbol{\alpha}\mathbf{F}, \qquad \forall\ \boldsymbol{\alpha} \in \mathbb{F}_q^N.$$

For the code $\mathbf{C}$, we define

$$R_{\boldsymbol{\alpha}}(\mathbf{C}) \triangleq \frac{1}{L} \cdot \sum\nolimits_{i=1}^N \log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}|, \qquad \forall\ \boldsymbol{\alpha} \in \mathbb{F}_q^N, \quad (3)$$

where "log" denotes "$\log_q$", the logarithm with base q. Then, $R_{\boldsymbol{\alpha}}(\mathbf{C})$ is the total communication cost, normalized by the file length, from all source nodes to the receiver ρ by using the code $\mathbf{C}$ for computing the linear function $\boldsymbol{\alpha}\mathbf{F}$. The *(worst-case) communication rate* of the code $\mathbf{C}$ is defined by

$$R(\mathbf{C}) \triangleq \max\nolimits_{\boldsymbol{\alpha}\in\mathbb{F}_q^N} R_{\boldsymbol{\alpha}}(\mathbf{C}). \quad (4)$$

Consequently, the *optimal communication rate* for the model $(\mathbb{F}_q, N, M)$ is defined by

$$R(\mathbb{F}_q, N, M) \triangleq \inf\Big\{R(\mathbf{C}) :\ \mathbf{C} \text{ is an admissible } L\text{-dimensional code, where } L \text{ is an arbitrary positive integer.}\Big\}.$$

When the model $(\mathbb{F}_q, N, M)$ is clear from the context, we simply write the optimal rate $R(\mathbb{F}_q, N, M)$ as $R(M)$.

III. Lower Bound on the Optimal Communication Rate

In this section, we prove a general lower bound on the optimal communication rate $R(\mathbb{F}_q, N, M)$, which is applicable for an arbitrary finite field $\mathbb{F}_q$, an arbitrary number N of sources, and an arbitrary normalized cache size M. Fix $1 \leq m,\ n \leq N$. For $A \in \mathbb{F}_q^{m\times N}$ and $\theta \subseteq [N]$ with $|\theta| = n$, let $A[\theta]$ be the submatrix formed by the columns indexed by θ. Define

$$\widehat{\operatorname{rank}}_n(A) = \frac{1}{\binom{N}{n}} \sum\nolimits_{\substack{\theta\subseteq[N]\\|\theta|=n}} \operatorname{rank}(A[\theta]), \quad (5)$$

$$\widehat{\operatorname{rank}}_{m,n,N} = \max\nolimits_{A\in\mathbb{F}_q^{m\times N}} \widehat{\operatorname{rank}}_n(A). \quad (6)$$

In particular, $\widehat{\operatorname{rank}}_{1,1,N} = 1$ and $\widehat{\operatorname{rank}}_{N,N,N} = N$.

[1] In this paper, for any positive integer Z, we use $\mathbb{F}_q^Z$ to denote the set of all Z-dimensional row vectors over $\mathbb{F}_q$.

[2] In the paper, we use $\operatorname{Im}\varphi$ to stand for the image set of a function φ.

Theorem 1. *For every finite field $\mathbb{F}_q$, $N \geq 1$, and $M \geq 0$,*

$$R(\mathbb{F}_q, N, M) \geq \max_{1\leq m,\, n\leq N} \frac{N}{mn}\big(\widehat{\text{rank}}_{m,n,N} - (N-n)M\big).$$

Proof. See Appendix A. □

IV. Code Construction and Upper Bound on the Optimal Communication Rate

In this section, we focus on the model $(\mathbb{F}_2, N, M)$ over the binary field $\mathbb{F}_2$. We construct a class of codes, referred to as *parity-aided-placement (PAP) codes*. With this code construction, we prove an upper bound on the optimal communication rate $R(\mathbb{F}_2, N, M)$.

A. Three Elementary Codes

We first present three elementary codes: an *uncached code*, a *full-placement code*, and an *all-but-one parity (ABO) code*, which will be combined to construct a PAP code.

• **Uncached Code:** Consider the model $(\mathbb{F}_2, N, M = 0)$, where N is an arbitrary positive integer and $M = 0$ indicates that no cached content is available at any source node. For each positive integer L, we define an L-dimensional *uncached code* $\mathbf{C} = \big\{\varphi_i, \phi_{i,\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} : i \in [N],\ \boldsymbol{\alpha} \in \mathbb{F}_2^N\big\}$ for the model $(\mathbb{F}_2, N, 0)$ as follows:

- no cached content is stored at any source node, i.e., the caching function φ_i at every source σ_i satisfies $\varphi_i(\mathbf{F}) = \langle\,\rangle$ for $i \in [N]$, where $\langle\,\rangle$ denotes the empty vector;
- the encoding function $\phi_{i,\boldsymbol{\alpha}}$ for each coefficient vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$ at every source node σ_i is given by $\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big) = \alpha_i \boldsymbol{F}_i$;
- the decoding function $\psi_{\boldsymbol{\alpha}}$ for each $\boldsymbol{\alpha} \in \mathbb{F}_2^N$ at the receiver ρ is given by

$$\psi_{\boldsymbol{\alpha}}\Big(\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big) : i \in [N]\Big) = \sum\nolimits_{i=1}^{N} \alpha_i \boldsymbol{F}_i = \boldsymbol{\alpha}\mathbf{F}.$$

For an arbitrary $i \in [N]$ and an arbitrary coefficient vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$, we can readily see that

$$\log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| = \begin{cases} L, & \alpha_i \neq 0, \\ 0, & \alpha_i = 0. \end{cases}$$

This further implies that

$$\begin{aligned}\sum\nolimits_{i=1}^{N} \log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| &= \big|\{i \in [N] :\ \alpha_i \neq 0\}\big| L \\ &= |\boldsymbol{\alpha}| L, \end{aligned} \tag{7}$$

where $|\boldsymbol{\alpha}|$ denotes the Hamming weight of $\boldsymbol{\alpha}$, i.e., the number of nonzero components of $\boldsymbol{\alpha}$.

• **Full-Placement Code:** Consider the model $(\mathbb{F}_2, N, M = (N-1)/N)$, where N is an arbitrary positive integer. For each positive integer L such that N divides L, we define an L-dimensional *full-placement code* $\mathbf{C} = \big\{\varphi_i, \phi_{i,\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} : i \in [N],\ \boldsymbol{\alpha} \in \mathbb{F}_2^N\big\}$ for the model $(\mathbb{F}_2, N, (N-1)/N)$ as follows. First, we partition $[L]$ into N disjoint subsets $\mathcal{L}_1, \mathcal{L}_2, \cdots, \mathcal{L}_N$ of equal size, i.e.,

$$[L] = \bigcup_{i=1}^{N} \mathcal{L}_i, \quad \text{and} \quad |\mathcal{L}_i| = \frac{L}{N}, \quad \forall\ i \in [N]. \tag{8}$$

- the caching function φ_i for generating the cache content at every source node σ_i is given by

$$\varphi_i(\mathbf{F}) = \Big(F_{n,\ell} :\ n \in [N] \setminus \{i\},\ \ell \in \mathcal{L}_i\Big), \tag{9}$$

  where we note that $|\mathrm{Im}\,\varphi_i| = 2^{(N-1)|\mathcal{L}_i|} = 2^{(N-1)L/N}$;
- the encoding function $\phi_{i,\boldsymbol{\alpha}}$ for each coefficient vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$ at every source node σ_i is given by

$$\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big) = \left(\sum\nolimits_{n=1}^{N} \alpha_n F_{n,\ell} :\ \ell \in \mathcal{L}_i\right); \tag{10}$$

- the decoding function $\psi_{\boldsymbol{\alpha}}$ for each $\boldsymbol{\alpha} \in \mathbb{F}_2^N$ at the receiver ρ is given by

$$\begin{aligned}&\psi_{\boldsymbol{\alpha}}\Big(\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big) : i \in [N]\Big) \\ &= \left(\sum\nolimits_{n=1}^{N} \alpha_n F_{n,\ell} :\ \ell \in [L]\right) = \sum\nolimits_{n=1}^{N} \alpha_n \boldsymbol{F}_n.\end{aligned}$$

We can readily see from (8), (10), and the definition of the decoding function that the code $\mathbf{C}$ is admissible. Furthermore, for any coefficient vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$,

$$\sum\nolimits_{i=1}^{N} \log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| = \begin{cases} 0 & \text{if } |\boldsymbol{\alpha}| = 0, \\ L & \text{if } |\boldsymbol{\alpha}| > 0. \end{cases} \tag{11}$$

To show this, when $|\boldsymbol{\alpha}| = 0$, we have $\boldsymbol{\alpha} = \mathbf{0}$, and hence $\log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| = 0$ for all $i \in [N]$. This immediately implies that $\sum_{i=1}^{N} \log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| = 0$. When $|\boldsymbol{\alpha}| > 0$, by (8) and (10), we have $\log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| = \frac{L}{N}$ for all $i \in [N]$. Therefore, we obtain that $\sum_{i=1}^{N} \log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| = \sum_{i=1}^{N} \frac{L}{N} = L$.

• **All-But-One Parity Code:** Consider the model $(\mathbb{F}_2, N, M = 1/N)$, where $N \geq 2$ is an arbitrary positive integer. For each positive integer L such that N divides L, we define an L-dimensional *all-but-one (ABO) parity code* $\mathbf{C} = \big\{\varphi_i, \phi_{i,\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} : i \in [N],\ \boldsymbol{\alpha} \in \mathbb{F}_2^N\big\}$ for the model $(\mathbb{F}_2, N, 1/N)$ as follows. First, we partition $[L]$ into N disjoint subsets $\mathcal{L}_1, \mathcal{L}_2, \cdots, \mathcal{L}_N$ of equal size as in (8).

- The caching function φ_i for generating the cache content at every source node σ_i is given by

$$\varphi_i(\mathbf{F}) = \left(\sum\nolimits_{n\in[N]\setminus\{i\}} F_{n,\ell} :\ \ell \in \mathcal{L}_i\right), \tag{12}$$

  where we note that $\big|\mathrm{Im}\,\varphi_i\big| = 2^{|\mathcal{L}_i|} = 2^{L/N}$ by (8);
- the encoding function $\phi_{i,\boldsymbol{\alpha}}$ for each coefficient vector $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$ at every source node σ_i is given by

$$\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big) = \big(X_{i,\boldsymbol{\alpha}}^{(\ell)} :\ \ell \in [L]\big), \tag{13}$$

where

$$X_{i,\boldsymbol{\alpha}}^{(\ell)} \triangleq \tag{14}$$
$$\begin{cases} \alpha_i F_{i,\ell} & \text{if } |\boldsymbol{\alpha}| \le N/2, \\ \alpha_i F_{i,\ell} + \sum_{n\in[N]\setminus\{i\}} F_{n,\ell} & \text{if } |\boldsymbol{\alpha}| > N/2 \text{ and } \ell \in \mathcal{L}_i, \\ (1-\alpha_i)F_{i,\ell}, & \text{if } |\boldsymbol{\alpha}| > N/2 \text{ and } \ell \notin \mathcal{L}_i; \end{cases}$$

- the decoding function $\psi_{\boldsymbol{\alpha}}$ for each $\boldsymbol{\alpha} \in \mathbb{F}_2^N$ at the receiver ρ is given by

$$\psi_{\boldsymbol{\alpha}}(\phi_{i,\boldsymbol{\alpha}}(\boldsymbol{F}_i, \varphi_i(\mathbf{F})) : i \in [N]) = (\sum_{i=1}^N X_{i,\boldsymbol{\alpha}}^{(\ell)} : \ell \in [L]).$$

We next show that the code $\mathbf{C}$ is admissible, i.e., for each $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$ and each $\ell \in [L]$, $\sum_{i=1}^N X_{i,\boldsymbol{\alpha}}^{(\ell)} = \sum_{i=1}^N \alpha_i F_{i,\ell}$; and for any $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_N) \in \mathbb{F}_2^N$,

$$\sum_{i=1}^N \log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| \tag{15}$$
$$= \begin{cases} |\boldsymbol{\alpha}|L & \text{if } |\boldsymbol{\alpha}| \le N/2, \\ \left(N - |\boldsymbol{\alpha}| + \frac{|\boldsymbol{\alpha}|}{N}\right)L & \text{if } |\boldsymbol{\alpha}| > N/2. \end{cases}$$

Towards this end, we consider two cases below.

Case 1: $|\boldsymbol{\alpha}| \le N/2$.

For this case, by (14), we have $\sum_{i=1}^N X_{i,\boldsymbol{\alpha}}^{(\ell)} = \sum_{i=1}^N \alpha_i F_{i,\ell}$. Moreover, by (13) and (14), we can readily see that

$$\log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| = \begin{cases} L & \text{if } \alpha_i \neq 0, \\ 0 & \text{if } \alpha_i = 0. \end{cases}$$

This immediately implies that $\sum_{i=1}^N \log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| = |\boldsymbol{\alpha}|L$.

Case 2: $|\boldsymbol{\alpha}| > N/2$.

Let $\mathrm{I}(\ell) \in [N]$ denote the unique index such that $\ell \in \mathcal{L}_{\mathrm{I}(\ell)}$. By (14), we have

$$\begin{aligned} &\sum_{i=1}^N X_{i,\boldsymbol{\alpha}}^{(\ell)} = X_{\mathrm{I}(\ell),\boldsymbol{\alpha}}^{(\ell)} + \sum_{i\in[N]\setminus\{\mathrm{I}(\ell)\}} X_{i,\boldsymbol{\alpha}}^{(\ell)} \\ &= \alpha_{\mathrm{I}(\ell)} F_{\mathrm{I}(\ell),\ell} + \sum_{n\in[N]\setminus\{\mathrm{I}(\ell)\}} F_{n,\ell} + \sum_{i\in[N]\setminus\{\mathrm{I}(\ell)\}} (1-\alpha_i)F_{i,\ell} \\ &= \alpha_{\mathrm{I}(\ell)} F_{\mathrm{I}(\ell),\ell} + \sum_{i\in[N]\setminus\{\mathrm{I}(\ell)\}} \alpha_i F_{i,\ell} \\ &= \sum_{i=1}^N \alpha_i F_{i,\ell}, \end{aligned}$$

where the third equality follows from $1 + (1-\alpha_i) = \alpha_i$ over $\mathbb{F}_2$ for every $i \in [N] \setminus \{\mathrm{I}(\ell)\}$. By (8), (13) and (14), we have

$$\log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| = \begin{cases} L/N & \text{if } \alpha_i \neq 0, \\ L & \text{if } \alpha_i = 0. \end{cases}$$

Therefore, we have

$$\begin{aligned} \sum_{i=1}^N \log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| &= |\boldsymbol{\alpha}|\frac{L}{N} + \big(N - |\boldsymbol{\alpha}|\big)L \\ &= \left(N - |\boldsymbol{\alpha}| + \frac{|\boldsymbol{\alpha}|}{N}\right)L. \end{aligned}$$

B. Parity-Aided-Placement Code

In this subsection, we consider the model $(\mathbb{F}_2, N, M)$ with $N \ge 2$ and construct a parity-aided-placement (PAP) code for this model by concatenating an uncached code, a full-placement code, and an ABO parity code.

We first present some notation. Let L_0, L_1, L_2 be arbitrary nonnegative integers such that N divides both L_1 and L_2 and

$$(N-1)L_1 + L_2 \le NM(L_0 + L_1 + L_2). \tag{16}$$

Let $L \triangleq L_0 + L_1 + L_2 > 0$. As will become clear below, the condition in (16) guarantees that the cache capacity constraint is satisfied. Further, we let

$$\lambda_0 \triangleq \frac{L_0}{L}, \quad \lambda_1 \triangleq \frac{L_1}{L}, \quad \text{and} \quad \lambda_2 \triangleq \frac{L_2}{L}. \tag{17}$$

Correspondingly, the inequality (16) can be rewritten as

$$(N-1)\lambda_1 + \lambda_2 \le NM. \tag{18}$$

For every $i \in [N]$, partition the file as $\boldsymbol{F}_i = [\boldsymbol{F}_i^{(0)}\ \boldsymbol{F}_i^{(1)}\ \boldsymbol{F}_i^{(2)}]$, where $\boldsymbol{F}_i^{(j)} \in \mathbb{F}_2^{L_j}$ for $j = 0, 1, 2$. Let $\mathbf{F}^{(j)} \in \mathbb{F}_2^{N\times L_j}$ be the matrix whose ith row is $\boldsymbol{F}_i^{(j)}$.

Furthermore, we let $\mathbf{C}^{(0)}$, $\mathbf{C}^{(1)}$, and $\mathbf{C}^{(2)}$ denote an L_0-dimensional uncached code for the model $(\mathbb{F}_2, N, 0)$, an L_1-dimensional full-placement code for the model $\big(\mathbb{F}_2, N, (N-1)/N\big)$, and an L_2-dimensional ABO parity code for the model $(\mathbb{F}_2, N, 1/N)$, respectively. Specifically, we write

$$\mathbf{C}^{(j)} = \big\{\varphi_i^{(j)}, \phi_{i,\boldsymbol{\alpha}}^{(j)}, \psi_{\boldsymbol{\alpha}}^{(j)} : i \in [N],\ \boldsymbol{\alpha} \in \mathbb{F}_2^N\big\}, \quad j = 0, 1, 2.$$

Based on these three codes, we construct an L-dimensional PAP code $\mathbf{C} = \big\{\varphi_i, \phi_{i,\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} : i \in [N],\ \boldsymbol{\alpha} \in \mathbb{F}_2^N\big\}$ for the model $(\mathbb{F}_2, N, M)$ as follows:

- the caching function φ_i at every source node σ_i is given by $\varphi_i(\mathbf{F}) = \Big[\varphi_i^{(j)}(\mathbf{F}^{(j)}) :\ j = 0, 1, 2\Big]$;
- the encoding function $\phi_{i,\boldsymbol{\alpha}}$ for each coefficient vector $\boldsymbol{\alpha} \in \mathbb{F}_2^N$ at every source node σ_i is given by

$$\phi_{i,\boldsymbol{\alpha}}\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big) = \Big[\phi_{i,\boldsymbol{\alpha}}^{(j)}\big(\boldsymbol{F}_i^{(j)}, \varphi_i^{(j)}(\mathbf{F}^{(j)})\big) :\ j = 0, 1, 2\Big];$$

- the decoding function $\psi_{\boldsymbol{\alpha}}$ for each $\boldsymbol{\alpha} \in \mathbb{F}_2^N$ at the receiver ρ is given by

$$\begin{aligned} &\psi_{\boldsymbol{\alpha}}(\phi_{i,\boldsymbol{\alpha}}(\boldsymbol{F}_i, \varphi_i(\mathbf{F})) : i \in [N]) \\ &\quad = \big[\psi_{\boldsymbol{\alpha}}^{(j)}(\phi_{i,\boldsymbol{\alpha}}^{(j)}(\boldsymbol{F}_i^{(j)}, \varphi_i^{(j)}(\mathbf{F}^{(j)})) : i \in [N]) :\ j = 0, 1, 2\big]. \end{aligned}$$

Since the three elementary codes are admissible, we can readily see that the code $\mathbf{C}$ is also admissible. For every $i \in [N]$, the caching functions act on disjoint file blocks, so $|\operatorname{Im}\varphi_i| = \prod_{j=0}^2 |\operatorname{Im}\varphi_i^{(j)}| = 2^{[(N-1)L_1+L_2]/N} \le 2^{ML}$, where the inequality follows from (16).

In the following, we calculate the communication rate of the code $\mathbf{C}$. Consider an arbitrary coefficient vector $\boldsymbol{\alpha} \in \mathbb{F}_2^N$. For each $i \in [N]$, we readily see that $|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| = |\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}^{(0)}| \cdot |\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}^{(1)}| \cdot |\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}^{(2)}|$. It follows that

$$\sum_{i=1}^N \log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}| = \sum_{j=0}^2 \sum_{i=1}^N \log|\operatorname{Im}\phi_{i,\boldsymbol{\alpha}}^{(j)}|. \tag{19}$$

Substituting the elementary-code rates in (7), (11), and (15) into (19), and using (17), gives

$$R_{\boldsymbol{\alpha}}(\mathbf{C}) = \frac{1}{L}\sum_{i=1}^{N} \log|\mathrm{Im}\,\phi_{i,\boldsymbol{\alpha}}| \tag{20}$$
$$= \begin{cases} 0 & \text{if } |\boldsymbol{\alpha}| = 0, \\ \lambda_0|\boldsymbol{\alpha}| + \lambda_1 + \lambda_2|\boldsymbol{\alpha}| & \text{if } 1 \le |\boldsymbol{\alpha}| \le N/2, \\ \lambda_0|\boldsymbol{\alpha}| + \lambda_1 + \lambda_2\big(N - |\boldsymbol{\alpha}| + \frac{|\boldsymbol{\alpha}|}{N}\big) & \text{if } |\boldsymbol{\alpha}| > N/2. \end{cases}$$

We readily see that $R_{\boldsymbol{\alpha}}(\mathbf{C})$ depends only on the Hamming weight $|\boldsymbol{\alpha}|$. For notational simplicity, we write $w \triangleq |\boldsymbol{\alpha}|$. Further, we note that

$$g_N(w) \triangleq \min\Big\{w, N - w + \frac{w}{N}\Big\} \tag{21}$$
$$= \begin{cases} w & \text{if } 0 \le w \le N/2, \\ N - w + \frac{w}{N} & \text{if } w > N/2. \end{cases}$$

With this, we can rewrite $R_{\boldsymbol{\alpha}}(\mathbf{C})$ as

$$R_{\boldsymbol{\alpha}}(\mathbf{C}) = \begin{cases} 0 & \text{if } w = 0, \\ \lambda_0 w + \lambda_1 + \lambda_2 g_N(w) & \text{if } 1 \le w \le N. \end{cases}$$

Consequently, we have

$$R(\mathbf{C}) = \max_{\boldsymbol{\alpha}\in\mathbb{F}_2^N} R_{\boldsymbol{\alpha}}(\mathbf{C})$$
$$= \max_{w\in[N]} \big\{\lambda_0 w + \lambda_1 + \lambda_2 g_N(w)\big\}. \tag{22}$$

We note that L_0, L_1, L_2 can be arbitrarily chosen such that N divides both L_1 and L_2 and (16) holds. Thus, by (17), any nonnegative $(\lambda_0, \lambda_1, \lambda_2)$ satisfying $\lambda_0 + \lambda_1 + \lambda_2 = 1$ and (18) can be approximated arbitrarily closely. Together with (22), this immediately gives the following theorem.

Theorem 2. *Consider the model $(\mathbb{F}_2, N, M)$ with $N \ge 2$. Let $R_{\mathrm{PAP}}(N, M) \triangleq \min_{\lambda_0,\lambda_1,\lambda_2} \max_{w\in[N]} \{\lambda_0 w + \lambda_1 + \lambda_2 g_N(w)\}$, where the minimum is subject to $\lambda_0, \lambda_1, \lambda_2 \ge 0, \lambda_0 + \lambda_1 + \lambda_2 = 1, (N-1)\lambda_1 + \lambda_2 \le NM$. Then, $R(\mathbb{F}_2, N, M) \le R_{\mathrm{PAP}}(N, M)$.*

The following theorem gives an explicit expression for $R_{\mathrm{PAP}}(N, M)$, whose proof is given in Appendix B.

Theorem 3. *For the model $(\mathbb{F}_2, N, M)$ with $N \ge 3$, $R_{\mathrm{PAP}}(N, M)$ is given as follows.*

- *If N is odd, $R_{\mathrm{PAP}}(N, M)$ is*

$$\begin{cases} N - N(N-1)M & \text{if } M \in [0, \frac{1}{2N-1}], \\ 1 + \frac{(N-1)^2(N-1-NM)}{2N^2-4N+1} & \text{if } M \in [\frac{1}{2N-1}, \frac{N-1}{N}], \\ 1 & \text{if } M \ge \frac{N-1}{N}. \end{cases}$$

- *If N is even, $R_{\mathrm{PAP}}(N, M)$ is*

$$\begin{cases} N - N(N-1)M & \text{if } M \in [0, \frac{1}{2N-1}], \\ \frac{N}{2} + 1 - \frac{3N-2}{2}M & \text{if } M \in [\frac{1}{2N-1}, \frac{2}{3N-2}], \\ \frac{N(3N+1)-(3N-2)NM}{2(3N-1)} & \text{if } M \in [\frac{2}{3N-2}, \frac{N-1}{N}], \\ 1 & \text{if } M \ge \frac{N-1}{N}. \end{cases}$$

Combining this achievable rate with the converse gives the following exact characterization.

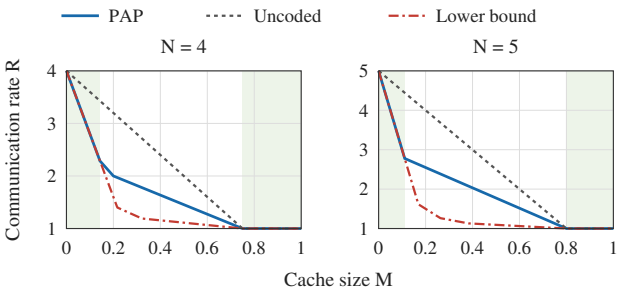


Fig. 2. Binary memory–rate tradeoffs for $N = 4, 5$. Shaded intervals mark the exact regimes in Corollary 1. The uncoded baseline uses only full and uncached coordinates.

Corollary 1. *For every $N \ge 3$,*

$$R(\mathbb{F}_2, N, M) = \begin{cases} N - N(N-1)M & \text{if } 0 \le M \le \frac{1}{2N-1}, \\ 1 & \text{if } M \ge \frac{N-1}{N}. \end{cases}$$

V. Numerical Results and Discussion

Figure 2 compares the PAP rate in Theorem 3 with the converse in Theorem 1 for $N = 4, 5$. The maximum average ranks are evaluated by enumerating binary query row spaces for each pair (m, n). We also consider an uncoded baseline using only full and uncached coordinates, which achieves the worst-case rate $\max\{1, N(1-M)\}$.

For $0 \le M \le 1/(2N-1)$, PAP meets the converse and reduces the rate with increasing cache size at $N-1$ times the slope magnitude of the baseline. In this regime, full-support requests determine the worst-case rate. An ABO coordinate serves such a request with one transmitted bit using one cached parity and the holder's local symbol, whereas a full coordinate requires caching $N-1$ symbols to achieve the same delivery cost. For $M \ge (N-1)/N$, both schemes achieve the optimal rate of one.

For $N = 4, 5$, PAP improves upon the baseline at intermediate cache sizes but does not meet the lower bound. At $N = 5$ and $M = 0.5$, for example, the PAP rate is 1.774, compared with 2.5 for the baseline and a lower bound of 1.094. The optimal memory–rate tradeoff in this intermediate regime remains open.

VI. Conclusion

We formulated a cache-aided linear computation model in which each source retains its own file and caches additional information before the requested linear combination is known. We established a converse bound over arbitrary finite fields by considering multiple requests under the same cache placement. For binary files, we proposed parity-aided placement (PAP) and obtained a closed-form worst-case rate for every $N \ge 3$ by optimizing its placement parameters. The resulting achievable rate meets the converse for $0 \le M \le 1/(2N-1)$ and $M \ge (N-1)/N$, establishing the optimal memory–rate tradeoff in these two regimes.

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APPENDIX A
PROOF OF THE CONVERSE

Proof: Consider an admissible L-dimensional code $\mathbf{C} = \big\{\varphi_i, \phi_{i,\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} : i \in [N],\ \boldsymbol{\alpha} \in \mathbb{F}_q^N\big\}$ for the model $(\mathbb{F}_q, N, M)$. We let m, n be arbitrary positive integers with $1 \le m,\, n \le N$. Further, we let $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \cdots, \boldsymbol{\alpha}_m$ be m arbitrary coefficient vectors. We consider the communication rate of the code $\mathbf{C}$

$$\begin{aligned} R(\mathbf{C}) &= \max_{\boldsymbol{\alpha} \in \mathbb{F}_q^N} R_{\boldsymbol{\alpha}}(\mathbf{C}) \\ &\ge \frac{1}{m} \cdot \sum_{k=1}^{m} R_{\boldsymbol{\alpha}_k}(\mathbf{C}) \\ &= \frac{1}{m} \cdot \sum_{k=1}^{m} \frac{1}{L} \cdot \sum_{i=1}^{N} \log |\mathrm{Im}\, \phi_{i,\boldsymbol{\alpha}_k}| \\ &\ge \frac{1}{mL} \cdot \sum_{k=1}^{m} \sum_{i=1}^{N} H\Big(\phi_{i,\boldsymbol{\alpha}_k}\big(\boldsymbol{F}_i,\, \varphi_i(\mathbf{F})\big)\Big), \end{aligned} \tag{23}$$

where $H(\,\cdot\,)$ denotes the entropy function computed with logarithm base q.

For each $i \in [N]$ and $k \in [m]$, let $X_{i,\boldsymbol{\alpha}_k}$ be the encoded message from the source node σ_i to the receiver ρ associated with the coefficient vector $\boldsymbol{\alpha}_k$, i.e.,

$$X_{i,\boldsymbol{\alpha}_k} \triangleq \phi_{i,\boldsymbol{\alpha}_k}\big(\boldsymbol{F}_i,\, \varphi_i(\mathbf{F})\big). \tag{24}$$

With this, we further let

$$\boldsymbol{X}_i \triangleq \big(X_{i,\boldsymbol{\alpha}_1}, X_{i,\boldsymbol{\alpha}_2}, \cdots, X_{i,\boldsymbol{\alpha}_m}\big), \qquad \forall\ i \in [N]. \tag{25}$$

Continuing from (23), we have

$$R(\mathbf{C}) \ge \frac{1}{mL} \cdot \sum_{i=1}^{N} \sum_{k=1}^{m} H(X_{i,\boldsymbol{\alpha}_k}) \ge \frac{1}{mL} \cdot \sum_{i=1}^{N} H(\boldsymbol{X}_i), \tag{26}$$

where the last inequality in (26) follows from the subadditivity of entropy.

Next, we lower bound $\sum_{i=1}^{N} H(\boldsymbol{X}_i)$. For notational simplicity, we let

$$\boldsymbol{X}_\theta \triangleq (\boldsymbol{X}_i :\ i \in \theta), \qquad \forall\ \theta \subseteq [N].$$

In particular, we let $\mathbf{X} \triangleq (\boldsymbol{X}_i :\ i \in [N])$. Then, we have

$$\begin{aligned} \sum_{i=1}^{N} H(\boldsymbol{X}_i) &= \frac{1}{\binom{N-1}{n-1}} \cdot \sum_{\theta \subseteq [N] \text{ s.t. } |\theta| = n} \sum_{i \in \theta} H(\boldsymbol{X}_i) && (27) \\ &\ge \frac{1}{\binom{N-1}{n-1}} \cdot \sum_{\theta \subseteq [N] \text{ s.t. } |\theta| = n} H(\boldsymbol{X}_\theta), && (28) \end{aligned}$$

where the equality (27) follows because for any given $i \in [N]$, the number of subsets $\theta \subseteq [N]$ satisfying $|\theta| = n$ and $i \in \theta$ is equal to $\binom{N-1}{n-1}$, and the inequality (28) follows from the subadditivity of entropy.

Furthermore, we consider $H(\boldsymbol{X}_\theta)$ for a subset $\theta \subseteq [N]$ with $|\theta| = n$. For the file matrix $\mathbf{F} \in \mathbb{F}_q^{N \times L}$, we let $\boldsymbol{F}_\theta$ be the submatrix of $\mathbf{F}$ containing the ith row if $i \in \theta$, and let $\boldsymbol{F}_{\theta^{\mathrm{c}}}$ be the submatrix of $\mathbf{F}$ containing the ith row if $i \in \theta^{\mathrm{c}} \triangleq [N] \setminus \theta$. Then,

$$H(\boldsymbol{X}_\theta) \ge H(\boldsymbol{X}_\theta \mid \boldsymbol{F}_{\theta^{\mathrm{c}}}) = H(\mathbf{X} \mid \boldsymbol{F}_{\theta^{\mathrm{c}}}) - H(\boldsymbol{X}_{\theta^{\mathrm{c}}} \mid \boldsymbol{X}_\theta, \boldsymbol{F}_{\theta^{\mathrm{c}}}), \tag{29}$$

where the equality in (29) follows from the chain rule of entropy.

We first consider $H(\mathbf{X} \mid \boldsymbol{F}_{\theta^{\mathrm{c}}})$ in (29). Note that for each coefficient vector $\boldsymbol{\alpha}_k$ with $k \in [m]$, the code $\mathbf{C}$ can compute the linear function $\boldsymbol{\alpha}_k \mathbf{F}$ with zero error, i.e.,

$$\begin{aligned} &\psi_{\boldsymbol{\alpha}_k}\big(X_{i,\boldsymbol{\alpha}_k},\, i \in [N]\big) \\ &= \psi_{\boldsymbol{\alpha}_k}\Big(\phi_{i,\boldsymbol{\alpha}_k}(\boldsymbol{F}_i, \varphi_i(\mathbf{F})) : i \in [N]\Big) \\ &= \boldsymbol{\alpha}_k \mathbf{F}, \quad \forall k \in [m], \end{aligned} \tag{30}$$

where the first equality in (30) follows from (24). We let

$$A \triangleq \begin{bmatrix} \boldsymbol{\alpha}_1 \\ \boldsymbol{\alpha}_2 \\ \vdots \\ \boldsymbol{\alpha}_m \end{bmatrix} \in \mathbb{F}_q^{m \times N}. \tag{31}$$

By (30), we can see that the matrix $A\mathbf{F}$ is a function of

$$\mathbf{X} = (\boldsymbol{X}_1, \boldsymbol{X}_2, \cdots, \boldsymbol{X}_N) = \big((X_{i,\boldsymbol{\alpha}_k} :\ k \in [m]) :\ i \in [N]\big).$$

With this, we further have

$$\begin{aligned} H(\mathbf{X} \,|\, \boldsymbol{F}_{\theta^{\mathrm{c}}}) &\ge H(A\mathbf{F} \,|\, \boldsymbol{F}_{\theta^{\mathrm{c}}}) \\ &= H\big(A[\theta] \cdot \boldsymbol{F}_\theta + A[\theta^{\mathrm{c}}] \cdot \boldsymbol{F}_{\theta^{\mathrm{c}}} \,\big|\, \boldsymbol{F}_{\theta^{\mathrm{c}}}\big) \\ &= H\big(A[\theta] \cdot \boldsymbol{F}_\theta \big|\, \boldsymbol{F}_{\theta^{\mathrm{c}}}\big) \\ &= H(A[\theta] \cdot \boldsymbol{F}_\theta) && (32) \\ &= L \cdot \mathrm{rank}(A[\theta]), && (33) \end{aligned}$$

where the two equalities (32) and (33) follow because the files $\boldsymbol{F}_1, \boldsymbol{F}_2, \cdots, \boldsymbol{F}_N$ are mutually independent and uniformly distributed over $\mathbb{F}_q^L$.

On the other hand, we consider $H(\boldsymbol{X}_{\theta^{\mathrm{c}}} \mid \boldsymbol{X}_\theta, \boldsymbol{F}_{\theta^{\mathrm{c}}})$ in (29). To be specific, it is easy to see that

$$\begin{aligned} &H(\boldsymbol{X}_{\theta^{\mathrm{c}}} \mid \boldsymbol{X}_\theta, \boldsymbol{F}_{\theta^{\mathrm{c}}}) \\ &\le H(\boldsymbol{X}_{\theta^{\mathrm{c}}} \mid \boldsymbol{F}_{\theta^{\mathrm{c}}}) \\ &\le \sum_{i \in \theta^{\mathrm{c}}} H(\boldsymbol{X}_i \mid \boldsymbol{F}_{\theta^{\mathrm{c}}}) \\ &\le \sum_{i \in \theta^{\mathrm{c}}} H(\boldsymbol{X}_i \mid \boldsymbol{F}_i). \end{aligned} \tag{34}$$

We further note that for each $i \in \theta^c$, the random vector $\boldsymbol{X}_i = \big(X_{i,\boldsymbol{\alpha}_1}, X_{i,\boldsymbol{\alpha}_2}, \cdots, X_{i,\boldsymbol{\alpha}_m}\big)$ is a deterministic function of $\big(\boldsymbol{F}_i, \varphi_i(\mathbf{F})\big)$ (cf. (24)), which implies that $H(\boldsymbol{X}_i \mid \boldsymbol{F}_i) \le H\big(\varphi_i(\mathbf{F}) \mid \boldsymbol{F}_i\big)$. Continuing from (34), we obtain that

$$\begin{aligned} H(\boldsymbol{X}_{\theta^{\mathrm{c}}} \mid \boldsymbol{X}_\theta, \boldsymbol{F}_{\theta^{\mathrm{c}}}) &\le \sum_{i \in \theta^{\mathrm{c}}} H(\boldsymbol{X}_i \mid \boldsymbol{F}_i) \\ &\le \sum_{i \in \theta^{\mathrm{c}}} H\big(\varphi_i(\mathbf{F}) \,\big|\, \boldsymbol{F}_i\big) \\ &\le \sum_{i \in \theta^{\mathrm{c}}} H\big(\varphi_i(\mathbf{F})\big) \\ &\le \sum_{i \in \theta^{\mathrm{c}}} ML = (N-n)ML, \end{aligned} \tag{35}$$

where the inequality (35) follows because $H\big(\varphi_i(\mathbf{F})\big) \le \log|\mathrm{Im}\,\varphi_i| \le ML$ for each $i \in [N]$.

For each subset $\theta \subseteq [N]$ with $|\theta| = n$, by (29), (33), and (35), we obtain

$$H(\boldsymbol{X}_\theta) \ge L \cdot \big(\mathrm{rank}(A[\theta]) - (N-n)M\big). \tag{36}$$

Combining (36) with (28), we have

$$\begin{aligned}
&\sum_{i=1}^{N} H(\boldsymbol{X}_i) \\
&\ge \frac{1}{\binom{N-1}{n-1}} \sum_{\theta\subseteq[N]:\,|\theta|=n} L \cdot \big(\mathrm{rank}(A[\theta]) - (N-n)M\big) \\
&= \frac{L}{\binom{N-1}{n-1}} \cdot \binom{N}{n} \cdot \left[\begin{array}{l} \dfrac{1}{\binom{N}{n}} \displaystyle\sum_{\theta\subseteq[N]:\,|\theta|=n} \mathrm{rank}(A[\theta]) \\ -\dfrac{1}{\binom{N}{n}} \displaystyle\sum_{\theta\subseteq[N]:|\theta|=n} (N-n)M \end{array} \right] \\
&= \frac{LN}{n} \cdot \Big[\widehat{\mathrm{rank}}_n(A) - (N-n)M\Big], \qquad (37)
\end{aligned}$$

where the equality (37) follows from $\binom{N}{n}/\binom{N-1}{n-1} = N/n$, the definition of $\widehat{\mathrm{rank}}_n(A)$ (cf. (5)), and

$$\begin{aligned}
&\frac{1}{\binom{N}{n}} \sum_{\theta\subseteq[N]:|\theta|=n} (N-n)M \\
&= \frac{1}{\binom{N}{n}} \binom{N}{n} (N-n)M \\
&= (N-n)M.
\end{aligned}$$

Continuing from (26), by (37) we have

$$R(\mathbf{C}) \ge \frac{N}{mn} \cdot \big(\widehat{\mathrm{rank}}_n(A) - (N-n)M\big). \tag{38}$$

We note that the inequality (38) holds for an arbitrary matrix $A \in \mathbb{F}_q^{m\times N}$, implying that

$$\begin{aligned}
R(\mathbf{C}) &\ge \max_{\text{all matrices } A\,\in\,\mathbb{F}_q^{m\times N}} \frac{N}{mn} \cdot \big(\widehat{\mathrm{rank}}_n(A) - (N-n)M\big) \\
&= \frac{N}{mn} \cdot \Big(\widehat{\mathrm{rank}}_{m,n,N} - (N-n)M\Big). \qquad (39)
\end{aligned}$$

Furthermore, the inequality (39) is true for arbitrary positive integers $1 \le m,\, n \le N$. As such, we have

$$R(\mathbf{C}) \ge \max_{1\le m,\,n\le N} \frac{N}{mn} \cdot \Big(\widehat{\mathrm{rank}}_{m,n,N} - (N-n)M\Big). \tag{40}$$

We finally note that the above lower bound (40) is valid for any positive integer L and any admissible L-dimensional code $\mathbf{C}$ for the model $(\mathbb{F}_q, N, M)$. The theorem is proved. ∎

Appendix B
Detailed Derivation of the PAP Minimax Solution

For $N = 2$, ABO placement coincides with full placement because the cached parity contains only the other source's symbol. Combining uncached and full-placement coordinates with full-placement fraction $\lambda_1 = \min\{2M, 1\}$ achieves worst-case rate $2 - \lambda_1$. The $m = n = 1$ and $m = n = 2$ instances of (1) give the matching lower bound, so $R(\mathbb{F}_2, 2, M) = R_{\mathrm{PAP}}(2, M) = \max\{2 - 2M, 1\}$ for all $M \ge 0$.

Proof of Theorem 3. For a nonzero request of weight w, write

$$r_{\mathrm{PAP}}(w; \lambda_1, \lambda_2) \triangleq (1 - \lambda_1 - \lambda_2)w + \lambda_1 + \lambda_2 g_N(w).$$

This is the per-request rate in (20), with $\lambda_0 = 1 - \lambda_1 - \lambda_2$.

For $M \ge (N-1)/N$, choosing $(\lambda_1, \lambda_2) = (1, 0)$ makes every coordinate a full coordinate and achieves rate 1. Moreover, $r_{\mathrm{PAP}}(1; \lambda_1, \lambda_2) = 1$ for every feasible pair (λ_1, λ_2), so no PAP code can have worst-case rate below 1.

It remains to consider $0 \le M \le (N-1)/N$. Over this range, we first reduce the optimization to one parameter. The three coordinate types are ordered by their per-source cache costs: an uncached coordinate requires no cache, a parity coordinate requires $1/N$, and a full coordinate requires $(N-1)/N$. Moving upward in this order never increases any per-request cost: replacing an uncached coordinate by a parity coordinate changes its cost from w to $g_N(w) \le w$, while replacing a parity coordinate by a full coordinate changes its cost from $g_N(w) \ge 1$ to 1.

Consequently, any feasible pair with $\frac{(N-1)\lambda_1+\lambda_2}{N} < M$ can be modified to increase its cache usage without increasing the value of the objective function, until the cache constraint is met with equality. It is therefore sufficient to optimize over pairs satisfying $\frac{(N-1)\lambda_1+\lambda_2}{N} = M$. By (18),

$$(N-1)\lambda_1 + \lambda_2 = NM. \tag{41}$$

We eliminate $\lambda_2 = NM - (N-1)\lambda_1$ and write

$$r_{\mathrm{PAP}}(w; \lambda_1) \triangleq r_{\mathrm{PAP}}\big(w; \lambda_1, NM - (N-1)\lambda_1\big).$$

After this substitution, the nonnegativity constraints $\lambda_1 \ge 0$, $\lambda_2 \ge 0$, and $1 - \lambda_1 - \lambda_2 \ge 0$ reduce to the feasible interval

$$\max\left\{0, \frac{NM-1}{N-2}\right\} \le \lambda_1 \le \frac{NM}{N-1}. \tag{42}$$

The resulting one-dimensional optimization is

$$\min_{\lambda_1} \max_{w\in[N]} r_{\mathrm{PAP}}(w; \lambda_1),$$

where λ_1 ranges over the interval above. It remains to characterize and minimize the upper envelope of the functions $r_{\mathrm{PAP}}(w; \lambda_1)$.

We next determine which request sizes $w \in [N]$ can attain $\max_{u\in[N]} r_{\mathrm{PAP}}(u; \lambda_1)$. For a fixed w, substituting $\lambda_2 = NM - (N-1)\lambda_1$ into (20) gives

$$\begin{aligned}
r_{\mathrm{PAP}}(w; \lambda_1) &= \lambda_1 + [NM - (N-1)\lambda_1] g_N(w) \\
&\quad + [1 - NM + (N-2)\lambda_1] w \\
&= \big[w + NM(g_N(w) - w)\big] \\
&\quad + \lambda_1 \big[1 - (N-1)g_N(w) + (N-2)w\big].
\end{aligned}$$

Thus, each request size defines an affine function of λ_1, and the worst-case rate is their upper envelope. To determine where the maximum over w, i.e., $\max_{w\in[N]} r_{\mathrm{PAP}}(w; \lambda_1)$, is attained,

we separately consider $w \le N/2$, for which $g_N(w) = w$, and $w > N/2$, for which $g_N(w) = N - w + w/N$.

Requests with $w \le N/2$**.**

In this regime,

$$\begin{aligned} r_{\mathrm{PAP}}(w;\lambda_1) &= \lambda_1 + [NM - (N-1)\lambda_1]w \\ &\quad + [1 - NM + (N-2)\lambda_1]w \\ &= w + \lambda_1(1-w) \\ &= \lambda_1 + (1-\lambda_1)w. \end{aligned} \tag{43}$$

For $w \ge 2$, this rate is strictly decreasing in λ_1. Moreover, $\lambda_1 \le 1$ throughout the feasible interval, so the last expression is non-decreasing in w. Hence, the maximum over $w \le N/2$ is attained at

$$w_L \triangleq \left\lfloor \frac{N}{2} \right\rfloor.$$

Requests with $w > N/2$**.**

In this regime, a direct computation gives

$$\begin{aligned} r_{\mathrm{PAP}}(w;\lambda_1) &= w + M[N^2 - w(2N-1)] \\ &\quad + \lambda_1\Big[1 - N^2 + N + \tfrac{w(2N^2-4N+1)}{N}\Big]. \end{aligned}$$

To identify the maximum over $w > N/2$, collect the terms that are independent of w and proportional to w, respectively:

$$\begin{aligned} r_{\mathrm{PAP}}(w;\lambda_1) &= N^2M + \lambda_1(1 - N^2 + N) \\ &\quad + w\left[1 - (2N-1)M + \frac{2N^2-4N+1}{N}\lambda_1\right]. \end{aligned} \tag{44}$$

Define

$$\begin{aligned} A(\lambda_1) &\triangleq N^2M + \lambda_1(1 - N^2 + N), \\ G(\lambda_1) &\triangleq 1 - (2N-1)M + \frac{2N^2-4N+1}{N}\lambda_1. \end{aligned} \tag{45}$$

Then (44) becomes

$$r_{\mathrm{PAP}}(w;\lambda_1) = A(\lambda_1) + wG(\lambda_1).$$

For $w > N/2$, the smallest admissible integer request size is

$$w_H \triangleq \left\lfloor \frac{N}{2} \right\rfloor + 1.$$

By (44), $r_{\mathrm{PAP}}(w;\lambda_1)$ is affine in w over the integers from w_H to N. Hence,

$$\max_{w > N/2} r_{\mathrm{PAP}}(w;\lambda_1) = \max\left\{r_{\mathrm{PAP}}(w_H;\lambda_1), r_{\mathrm{PAP}}(N;\lambda_1)\right\}.$$

Combining this identity with the maximum over $w \le N/2$ gives

$$\begin{aligned} \max_{w \in [N]} r_{\mathrm{PAP}}(w;\lambda_1) = \max\big\{&r_{\mathrm{PAP}}(w_L;\lambda_1), \\ &r_{\mathrm{PAP}}(w_H;\lambda_1), r_{\mathrm{PAP}}(N;\lambda_1)\big\}. \end{aligned} \tag{46}$$

Having reduced the worst-case rate to three affine functions, we now minimize their upper envelope by comparing how these functions vary with λ_1. For $w > N/2$, define

$$\Gamma(w) \triangleq 1 - N^2 + N + \frac{w(2N^2-4N+1)}{N},$$

so that $\Gamma(w)$ is the slope of $r_{\mathrm{PAP}}(w;\lambda_1)$ with respect to λ_1.

From (43) and (44), the slopes of $r_{\mathrm{PAP}}(w_L;\lambda_1)$, $r_{\mathrm{PAP}}(w_H;\lambda_1)$, and $r_{\mathrm{PAP}}(N;\lambda_1)$ with respect to λ_1 are, respectively,

$$1 - w_L, \qquad \Gamma(w_H), \qquad \Gamma(N).$$

For $N \ge 3$,

$$1 - w_L \le 0, \qquad \Gamma(N) = (N-1)(N-2) > 0.$$

Thus, $r_{\mathrm{PAP}}(w_L;\lambda_1)$ is non-increasing and $r_{\mathrm{PAP}}(N;\lambda_1)$ is strictly increasing in λ_1.

The sign of $\Gamma(w_H)$ depends on whether N is odd or even, so we consider these two cases separately.

Odd N.

For odd N,

$$w_L = \frac{N-1}{2}, \qquad w_H = \frac{N+1}{2}.$$

The preceding envelope reduction leaves w_L, w_H, and N as the only candidate request sizes. We first show that the rate at w_L never exceeds that at w_H. Indeed,

$$g_N(w_L) = w_L, \qquad g_N(w_H) = w_L + \frac{N+1}{2N}.$$

Therefore, (20) gives, for every feasible pair (λ_1, λ_2),

$$\begin{aligned} &r_{\mathrm{PAP}}(w_H;\lambda_1,\lambda_2) - r_{\mathrm{PAP}}(w_L;\lambda_1,\lambda_2) \\ &\quad = \lambda_2\big(g_N(w_H) - g_N(w_L)\big) + (1 - \lambda_1 - \lambda_2)(w_H - w_L) \\ &\quad = \lambda_2\frac{N+1}{2N} + (1 - \lambda_1 - \lambda_2) \ge 0. \end{aligned}$$

Hence, after eliminating λ_2,

$$\max_{w \in [N]} r_{\mathrm{PAP}}(w;\lambda_1) = \max\left\{r_{\mathrm{PAP}}(w_H;\lambda_1), r_{\mathrm{PAP}}(N;\lambda_1)\right\}.$$

The slopes of $r_{\mathrm{PAP}}(w_H;\lambda_1)$ and $r_{\mathrm{PAP}}(N;\lambda_1)$ with respect to λ_1 are

$$\Gamma(w_H) = -\frac{N-1}{2N} < 0$$

and

$$\Gamma(N) = (N-1)(N-2) > 0,$$

respectively. Thus, $r_{\mathrm{PAP}}(w_H;\lambda_1)$ is strictly decreasing in λ_1, whereas $r_{\mathrm{PAP}}(N;\lambda_1)$ is strictly increasing. Their unique intersection is equivalently characterized by $G(\lambda_1) = 0$ and occurs at

$$\lambda_{1,\mathrm{bal}} \triangleq \frac{N\big((2N-1)M - 1\big)}{2N^2 - 4N + 1}. \tag{47}$$

The maximum of these two rates is minimized at $\lambda_{1,\mathrm{bal}}$ when the intersection lies in the feasible interval. If the intersection lies to the left or right of this interval, the minimum is attained at the corresponding endpoint.

Case O1: $0 \le M \le 1/(2N-1)$.

By (42), the lower endpoint of the feasible interval for λ_1 is

$$\max\left\{0, \frac{NM - 1}{N-2}\right\}.$$

Since

$$M \le \frac{1}{2N-1} < \frac{1}{N},$$

we have $NM - 1 < 0$, so this lower endpoint equals 0. Moreover, (47) gives

$$\lambda_{1,\mathrm{bal}} \le 0.$$

The intersection therefore lies at or to the left of the feasible interval. For every feasible λ_1, (42) gives

$$\lambda_1 \ge 0 \ge \lambda_{1,\mathrm{bal}}.$$

Since $r_{\mathrm{PAP}}(N;\lambda_1)$ is increasing, $r_{\mathrm{PAP}}(w_H;\lambda_1)$ is decreasing, and the two rates are equal at $\lambda_{1,\mathrm{bal}}$, we have

$$\begin{aligned} r_{\mathrm{PAP}}(N;\lambda_1) &\ge r_{\mathrm{PAP}}(N;\lambda_{1,\mathrm{bal}}) \\ &= r_{\mathrm{PAP}}(w_H;\lambda_{1,\mathrm{bal}}) \\ &\ge r_{\mathrm{PAP}}(w_H;\lambda_1). \end{aligned}$$

Thus, $r_{\mathrm{PAP}}(N;\lambda_1)$ determines the upper envelope throughout the feasible interval. Since this rate is strictly increasing, the upper envelope is minimized at $\lambda_1^\star = 0$. Substituting $\lambda_1^\star = 0$ into (41) gives $\lambda_2^\star = NM$, and

$$R_{\mathrm{PAP}}(N,M) = r_{\mathrm{PAP}}(N;0) = N - N(N-1)M.$$

Case O2: $1/(2N-1) < M \le (N-1)/N$.

In this range, (47) gives $\lambda_{1,\mathrm{bal}} > 0$. By (42), the feasible interval is

$$\max\left\{0, \frac{NM-1}{N-2}\right\} \le \lambda_1 \le \frac{NM}{N-1}.$$

Its upper endpoint is $NM/(N-1)$. The difference between this endpoint and $\lambda_{1,\mathrm{bal}}$ is

$$\frac{NM}{N-1} - \lambda_{1,\mathrm{bal}} = \frac{N(N-1-NM)}{(N-1)(2N^2-4N+1)} \ge 0.$$

Hence, $\lambda_{1,\mathrm{bal}}$ does not exceed the upper endpoint.

It remains to compare $\lambda_{1,\mathrm{bal}}$ with the lower endpoint. If $M \le 1/N$, then $NM - 1 \le 0$, so the lower endpoint is 0 and is strictly smaller than $\lambda_{1,\mathrm{bal}}$. If $M > 1/N$, then the lower endpoint is $(NM-1)/(N-2)$, and

$$\lambda_{1,\mathrm{bal}} - \frac{NM-1}{N-2} = \frac{(N-1)(N-1-NM)}{(N-2)(2N^2-4N+1)} \ge 0.$$

Since $\lambda_{1,\mathrm{bal}}$ lies in the feasible interval throughout Case O2, the upper envelope is minimized at

$$\lambda_1^\star = \lambda_{1,\mathrm{bal}} = \frac{N\big((2N-1)M-1\big)}{2N^2-4N+1}.$$

Substituting this value into (41) gives

$$\lambda_2^\star = \frac{N\big(N-1-NM\big)}{2N^2-4N+1}.$$

Therefore, the minimized worst-case rate is

$$\begin{aligned} R_{\mathrm{PAP}}(N,M) &= r_{\mathrm{PAP}}(w_H;\lambda_1^\star) \\ &= r_{\mathrm{PAP}}(N;\lambda_1^\star) \\ &= 1 + \frac{(N-1)^2}{2N^2-4N+1}\big(N-1-NM\big). \end{aligned}$$

At $M = (N-1)/N$, the optimal pair becomes $(\lambda_1^\star, \lambda_2^\star) = (1,0)$ and the rate becomes 1, consistent with the full-coordinate regime.

Even N.

For even N, assume $N \ge 4$. Then

$$w_L = \frac{N}{2}, \qquad w_H = \frac{N}{2} + 1.$$

By (46), only the rates corresponding to w_L, w_H, and N can determine the upper envelope. Their dependence on λ_1 is particularly simple. For $w_L = N/2$,

$$r_{\mathrm{PAP}}(w_L;\lambda_1) = \frac{N}{2} - \frac{N-2}{2}\lambda_1,$$

which is strictly decreasing in λ_1. For the other two request sizes,

$$\begin{aligned} \Gamma(w_H) &= \frac{(2N-1)(N-2)}{2N} > 0, \\ \Gamma(N) &= (N-1)(N-2) > 0. \end{aligned}$$

Hence, both $r_{\mathrm{PAP}}(w_H;\lambda_1)$ and $r_{\mathrm{PAP}}(N;\lambda_1)$ are strictly increasing in λ_1.

We first determine when $\lambda_1 = 0$ minimizes the upper envelope. By (42), this choice is feasible precisely when $M \le 1/N$.

If one of the two increasing rates attains the maximum at $\lambda_1 = 0$, then increasing λ_1 increases that rate and cannot reduce the upper envelope. We therefore compare the three rates at $\lambda_1 = 0$.

Substituting $\lambda_1 = 0$ and $\lambda_2 = NM$ into (20) gives

$$\begin{aligned} r_{\mathrm{PAP}}(w_L;0) &= \frac{N}{2}, \\ r_{\mathrm{PAP}}(w_H;0) &= \frac{N}{2} + 1 - \frac{3N-2}{2}M, \\ r_{\mathrm{PAP}}(N;0) &= N - N(N-1)M. \end{aligned} \tag{48}$$

To identify the largest of these rates, we compute

$$\begin{aligned} r_{\mathrm{PAP}}(N;0) - r_{\mathrm{PAP}}(w_H;0) &= \frac{N-2}{2}\big[1-(2N-1)M\big], \\ r_{\mathrm{PAP}}(w_H;0) - r_{\mathrm{PAP}}(w_L;0) &= 1 - \frac{3N-2}{2}M. \end{aligned} \tag{49}$$

The two differences change sign at $M = 1/(2N-1)$ and $M = 2/(3N-2)$, respectively. Since $1/(2N-1) < 2/(3N-2) < 1/N$, both transitions occur while $\lambda_1 = 0$ is feasible. These two thresholds give the following three ranges of M.

Case E1: $0 \le M \le 1/(2N-1)$.

Both differences in (49) are nonnegative, so (48) gives

$$r_{\mathrm{PAP}}(N;0) \ge r_{\mathrm{PAP}}(w_H;0) \ge r_{\mathrm{PAP}}(w_L;0).$$

Thus, the upper envelope at $\lambda_1 = 0$ equals $r_{\mathrm{PAP}}(N;0)$. For every feasible $\lambda_1 \ge 0$, the monotonicity of $r_{\mathrm{PAP}}(N;\lambda_1)$ gives

$$\max_{w\in[N]} r_{\mathrm{PAP}}(w;\lambda_1) \ge r_{\mathrm{PAP}}(N;\lambda_1) \ge r_{\mathrm{PAP}}(N;0).$$

Therefore, the upper envelope is minimized at $\lambda_1^\star = 0$. Equation (41) then gives $\lambda_2^\star = NM$, and

$$R_{\mathrm{PAP}}(N,M) = r_{\mathrm{PAP}}(N;0) = N - N(N-1)M.$$

Case E2: $1/(2N-1) < M \le 2/(3N-2)$.

The first difference in (49) is negative, whereas the second is nonnegative. Hence,

$$r_{\mathrm{PAP}}(w_H;0) \ge \max\left\{r_{\mathrm{PAP}}(w_L;0), r_{\mathrm{PAP}}(N;0)\right\},$$

so the upper envelope at $\lambda_1 = 0$ equals $r_{\mathrm{PAP}}(w_H;0)$. For every feasible $\lambda_1 \ge 0$, the monotonicity of $r_{\mathrm{PAP}}(w_H;\lambda_1)$ gives

$$\max_{w\in[N]} r_{\mathrm{PAP}}(w;\lambda_1) \ge r_{\mathrm{PAP}}(w_H;\lambda_1) \ge r_{\mathrm{PAP}}(w_H;0).$$

Therefore, the upper envelope is again minimized at $\lambda_1^\star = 0$. By (41), $\lambda_2^\star = NM$, and

$$R_{\mathrm{PAP}}(N,M) = r_{\mathrm{PAP}}(w_H;0) = \frac{N}{2} + 1 - \frac{3N-2}{2}M.$$

Case E3: $2/(3N-2) < M \le (N-1)/N$.

For $2/(3N-2) < M \le 1/N$, (49) shows that $r_{\mathrm{PAP}}(w_L;0)$ strictly exceeds both increasing rates at $\lambda_1 = 0$. Thus, a sufficiently small feasible increase in λ_1 decreases the upper envelope. For $M > 1/N$, $\lambda_1 = 0$ is infeasible. We therefore seek a minimizer with $\lambda_1 > 0$.

In this regime, the rate $r_{\mathrm{PAP}}(w_L;\lambda_1)$ decreases with λ_1, whereas $r_{\mathrm{PAP}}(w_H;\lambda_1)$ increases. Their unique intersection is

$$\lambda_{1,\mathrm{bal}}^{(\mathrm{e})} \triangleq \frac{N\big((3N-2)M-2\big)}{(3N-1)(N-2)}. \tag{50}$$

We first verify that this intersection lies in the feasible interval (42). The lower bound on M in Case E3 gives $\lambda_{1,\mathrm{bal}}^{(\mathrm{e})} > 0$, and

$$\frac{NM}{N-1} - \lambda_{1,\mathrm{bal}}^{(\mathrm{e})} = \frac{2N(N-1-NM)}{(N-1)(3N-1)(N-2)} \ge 0,$$

so the intersection does not exceed the upper endpoint. If $M \le 1/N$, the lower endpoint in (42) is 0 and is therefore smaller than $\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}$. If $M > 1/N$, then the lower endpoint is $(NM-1)/(N-2)$, and

$$\lambda_{1,\mathrm{bal}}^{(\mathrm{e})} - \frac{NM-1}{N-2} = \frac{N-1-NM}{(3N-1)(N-2)} \ge 0.$$

Thus, (50) is feasible throughout Case E3.

Since $r_{\mathrm{PAP}}(w_L;\lambda_1)$ decreases and $r_{\mathrm{PAP}}(w_H;\lambda_1)$ increases, their common value at the feasible intersection $\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}$ is a lower bound on the worst-case rate. To show that this lower bound is attained, it remains to verify that the third candidate rate, $r_{\mathrm{PAP}}(N;\lambda_1)$, does not exceed this common value at the balancing point. Substituting (50) into (45) gives

$$G(\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}) = -\frac{N(N-1-NM)}{(3N-1)(N-2)} \le 0.$$

By (44),

$$\begin{aligned} &r_{\mathrm{PAP}}(N;\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}) - r_{\mathrm{PAP}}(w_H;\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}) \\ &\quad = (N-w_H)G(\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}) \le 0. \end{aligned}$$

Together with (46), this shows that the upper envelope at $\lambda_{1,\mathrm{bal}}^{(\mathrm{e})}$ equals the common value of the rates for w_L and w_H. The preceding lower bound therefore proves that

$$\lambda_1^\star = \lambda_{1,\mathrm{bal}}^{(\mathrm{e})} = \frac{N\big((3N-2)M-2\big)}{(3N-1)(N-2)}.$$

Substituting this value into (41) gives

$$\lambda_2^\star = \frac{2N\big(N-1-NM\big)}{(3N-1)(N-2)}.$$

Finally,

$$\begin{aligned} R_{\mathrm{PAP}}(N,M) &= r_{\mathrm{PAP}}(w_L;\lambda_1^\star) \\ &= r_{\mathrm{PAP}}(w_H;\lambda_1^\star) \\ &= \frac{N(3N+1)-(3N-2)NM}{2(3N-1)}. \end{aligned}$$

At $M = (N-1)/N$, the optimal pair is $(\lambda_1^\star,\lambda_2^\star) = (1,0)$ and the rate is 1, matching the full-coordinate regime. □

Appendix C
Finite-Length Realization

Fix rational $\lambda_1,\lambda_2 \ge 0$ with $\lambda_1+\lambda_2 \le 1$ and $(N-1)\lambda_1+\lambda_2 \le NM$. Choose L so that $\lambda_1 L$, $\lambda_2 L$, $\lambda_1 L/N$, and $\lambda_2 L/N$ are integers. Partition $[L]$ into full, ABO, and uncached sets of the prescribed sizes. Assign exactly $\lambda_1 L/N$ full coordinates and $\lambda_2 L/N$ ABO coordinates to each source. Source i caches the $N-1$ nonlocal symbols at each of its assigned full coordinates and one ABO parity at each of its assigned ABO coordinates. Its cache image has size

$$2^{L[(N-1)\lambda_1+\lambda_2]/N}.$$

The cached symbols at a full coordinate are independent, and an ABO parity is a nonzero linear form. Different coordinates involve disjoint file symbols, so the stated image size is exact. The delivery rules in Section IV recover every requested coordinate. Each source's transmitted nonzero linear forms also involve disjoint symbols, so the logarithm of its message image size equals its transmitted bit count. Summing these counts gives (20).

For a real feasible pair (λ_1,λ_2), choose rational sequences $0 \le \lambda_{1,k} \le \lambda_1$ and $0 \le \lambda_{2,k} \le \lambda_2$ converging to these parameters. Then $\lambda_{1,k}+\lambda_{2,k} \le 1$ and $(N-1)\lambda_{1,k}+\lambda_{2,k} \le NM$, so every approximating pair is feasible. The maximum of the finitely many request rates in (20) is continuous. The rational constructions therefore approach the rate of (λ_1,λ_2) while respecting the same cache budget, proving asymptotic achievability.